

Moment Generating Function (MGF)

Statistics Reference Notes

1. Moment Generating Function (MGF)

Definition

The **Moment Generating Function (MGF)** of a random variable X is defined as:

$$M_X(t) = E(e^{tX})$$

where

- E = expectation
- t = real parameter
- X = random variable

The MGF is called a **moment generating function** because it can be used to obtain the **moments of a distribution**, such as:

- Mean
 - Variance
 - Higher-order moments
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Why MGF is Important

The derivatives of the MGF help compute statistical moments.

Mean (First Moment)

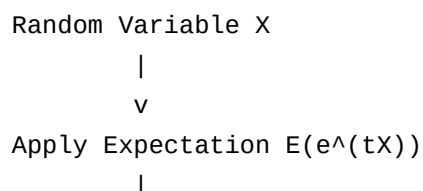
$$E(X) = M'_X(0)$$

Variance

$$\text{Var}(X) = M''_X(0) - [M'_X(0)]^2$$

Thus, the MGF provides a convenient method to compute **mean and variance** of probability distributions.

Concept Diagram



v
Moment Generating Function $M_X(t)$
|
v
Moments of Distribution
(mean, variance, etc.)

2. MGF for Continuous Random Variables

If X is a **continuous random variable** with probability density function $f(x)$, the MGF is:

$$M_X(t) = \int_{-\infty}^{\infty} e^{tx} f(x) dx$$

Interpretation

1. Multiply e^{tx} with the density function $f(x)$
 2. Integrate over the entire range of x
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3. MGF for Discrete Random Variables

If X is **discrete** with probability mass function $P(x)$, then:

$$M_X(t) = \sum e^{tx} P(x)$$

Steps

1. Multiply e^{tx} with probability $P(x)$
 2. Sum over all possible values of x
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4. Example of MGF

Consider a random variable:

x	$P(x)$
0	0.5
1	0.5

Step 1: Use formula

$$M_X(t) = \sum e^{tx} P(x)$$

Step 2: Substitute values

$$M_X(t) = e^{t(0)}(0.5) + e^{t(1)}(0.5)$$

Step 3: Simplify

$$M_X(t) = 0.5e^0 + 0.5e^t$$

Since $e^0 = 1$:

$$M_X(t) = 0.5 + 0.5e^t$$

5. MGF of Binomial Distribution

If $X \sim \text{Bin}(n, p)$, then the moment generating function is:

$$M_X(t) = (q + pe^t)^n$$

p = probability of success

$q = 1 - p$

Example

A coin is tossed **3 times**.

$n = 3$

$p = 0.5$

$q = 0.5$

MGF:

$$M_X(t) = (0.5 + 0.5e^t)^3$$

Sample Problem 1

Find the **MGF of a binomial distribution with $n = 4$ and $p = 0.3$** .

Solution

Given:

$n = 4$

$p = 0.3$

$q = 0.7$

Formula:

$$M_X(t) = (q + pe^t)^n$$

Substitute values:

$$M_X(t) = (0.7 + 0.3e^t)^4$$

6. MGF of Poisson Distribution

If $X \sim \text{Poisson}(\lambda)$, then:

$$M_X(t) = e^{\lambda(e^t - 1)}$$

λ = average number of occurrences

Example

Suppose average number of accidents per day is:

$$\lambda = 2$$

MGF:

$$M_X(t) = e^{2(e^t - 1)}$$

Sample Problem 2

Find the **MGF of Poisson distribution** when $\lambda = 5$.

Solution

Formula:

$$M_X(t) = e^{\lambda(e^t - 1)}$$

Substitute:

$$M_X(t) = e^{5(e^t - 1)}$$

7. MGF of Geometric Distribution

If X follows a **geometric distribution**, its MGF is:

$$M_X(t) = pe^t / (1 - qe^t)$$

p = probability of success

$$q = 1 - p$$

Example

Let p = 0.4, q = 0.6:

$$M_X(t) = 0.4e^t / (1 - 0.6e^t)$$

Sample Problem 3

Find the **MGF of geometric distribution** when p = 0.2.

Solution

Given:

$$p = 0.2$$

$$q = 0.8$$

Formula:

$$M_X(t) = pe^t / (1 - qe^t)$$

Substitute:

$$M_X(t) = 0.2e^t / (1 - 0.8e^t)$$

Quick Comparison of MGFs

Distribution	MGF
Binomial	$(q + pe^t)^n$
Poisson	$e^{\lambda(e^t - 1)}$
Geometric	$pe^t / (1 - qe^t)$

Basic Level Summary

Topics Covered

- ✓ Moment Generating Function
- ✓ MGF definition
- ✓ MGF for discrete variables
- ✓ MGF for continuous variables
- ✓ MGF of Binomial distribution
- ✓ MGF of Poisson distribution
- ✓ MGF of Geometric distribution